北京交通大學

信号与系统

School of Electronic
and Information Engineering

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2025.4

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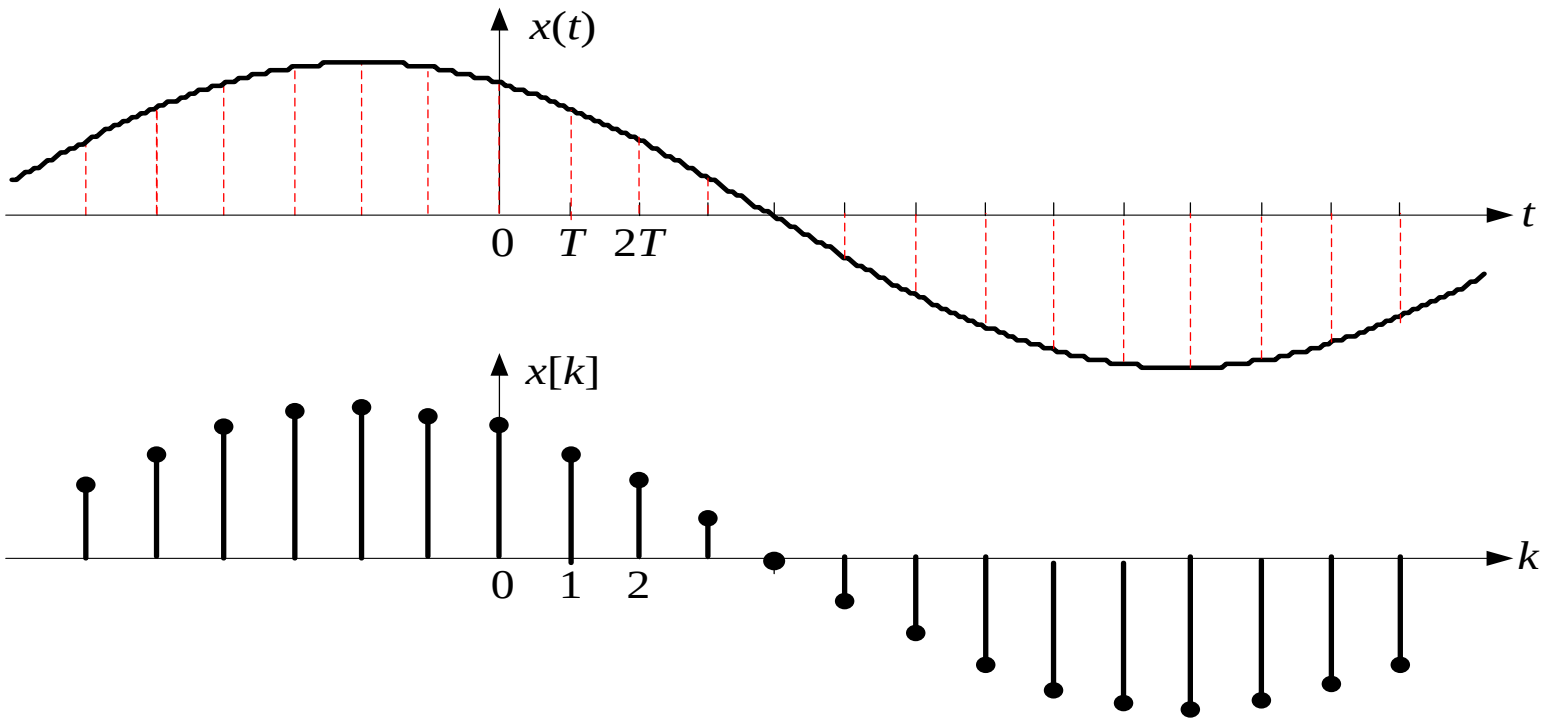
Frequency domain analysis of signals

- Frequency domain analysis of continuous periodic signals
- Frequency domain analysis of continuous aperiodic signals
- Frequency domain analysis of discrete periodic signals
- Frequency domain analysis of discrete aperiodic signals
- Time domain sampling of continuous-time signals

Time domain sampling of continuous-time signals

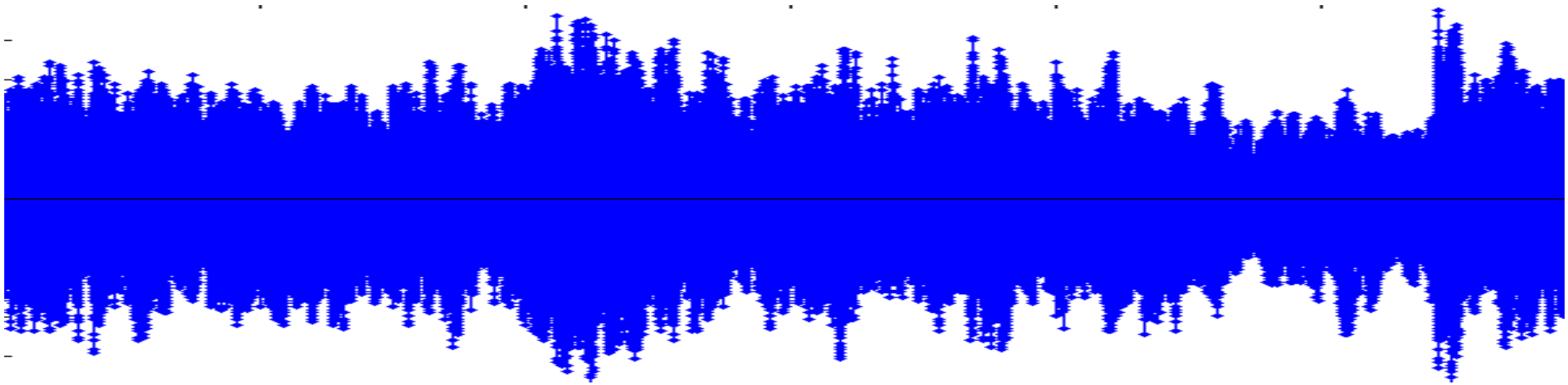
- ※ What is signal sampling?
- ※ Why do you want to sample?
- ※ How to sample signals
- ※ Theoretical analysis of signal sampling
- ※ Engineer application of sampling theorem

1. What is signal sampling?



$$x[k] = x(t)\big|_{t=kT}$$

1. What is signal sampling?



```
[x,Fs,Bits]=wavread('myheart');
play(x)
```

Fs=22,050; Bits=16

1. What is signal sampling?

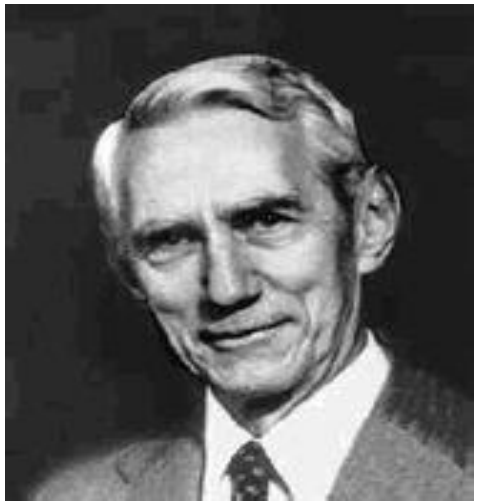
Who proposed signal sampling?

Harry Nyquist(1889 – 1976)

Nyquist, an American physicist, was born in Sweden in 1889. He made a great contribution to information theory. Immigrated to the United States in 1907 and entered the University of North Dakota in 1912. In 1917, he received his doctorate in physics from Yale University. From 1917 to 1934, he worked in AT&T Company, and then transferred to Bell Telephone Laboratory.

In 1927, Nyquist decided to sample a finite-time continuous signal with a certain bandwidth. In order not to cause "half-wave loss" in the original waveform, the sampling rate should be at least twice the highest frequency of the signal, which is the famous Nyquist sampling theorem.

Who proposed signal sampling?



Claude Elwood Shannon, 1916-2001, was born in Michigan, USA. In 1940, he received a doctorate in mathematics and a master's degree in electronic engineering from MIT. Shannon published two revolutionary articles, which laid the foundation of information theory.

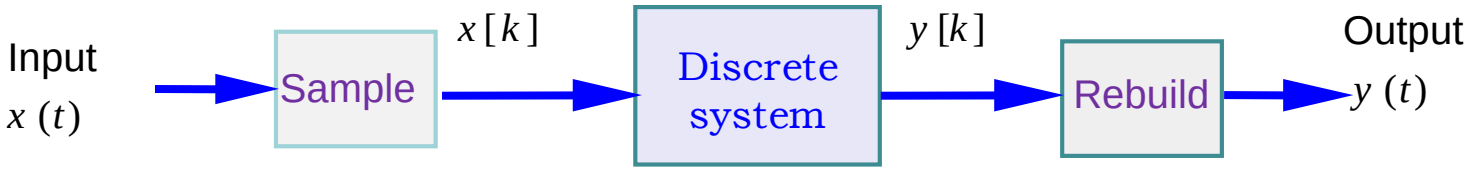
[l] C. E. Shannon, A mathematical theory of communication,1948.

[2] C. E. Shannon, Communication in the presence of noise, 1949.

V. A. Kotelnikov, Kotelnikov, was the first scientist who expounded the sampling theorem accurately and systematically and applied it to the field of communication engineering. He proved the sampling theory of low-pass signal and band-pass signal.

[1] V.A.Kotelnikov,On the transmission capacity of ‘ether‘ and wire in electro-communications,1933.

2. Why do you sample signals?



Using digital technology to process analog signals

Main advantages of discrete signals and systems:

※Good signal stability: the data is expressed in binary, which is less influenced by the outside world.

※High signal reliability: no loss in storage and anti-interference in transmission.

※Simple signal processing: signal compression, signal coding, signal encryption, etc.

※High accuracy of the system: the accuracy of the system can be improved by increasing the word length.

※Strong system flexibility: changing the coefficient of the system enables the system to complete different functions.

Main characteristics of digital signal

※Easy to store

※Easy transmission

※Easy to handle

Main characteristics of digital signal

Easy to store

Storage mode

Analog storage

tape

videotape

Camera film

Digital storage

DVD

hard disc

Sd card

Advantages of digital storage:

1. Large capacity
2. Good stability

Main characteristics of digital signal

Easy to store

Good stability

Old photos

Digital photos (spatially discrete)

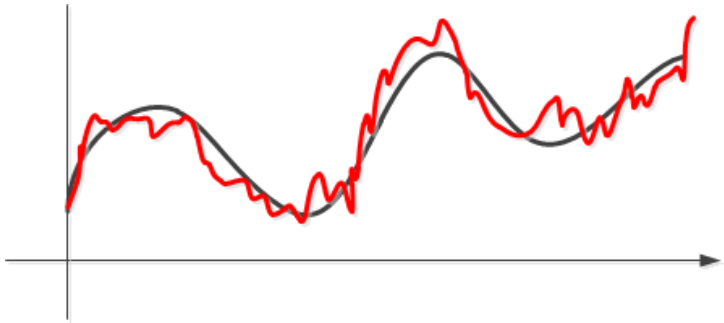
Durian Song
Tape quality

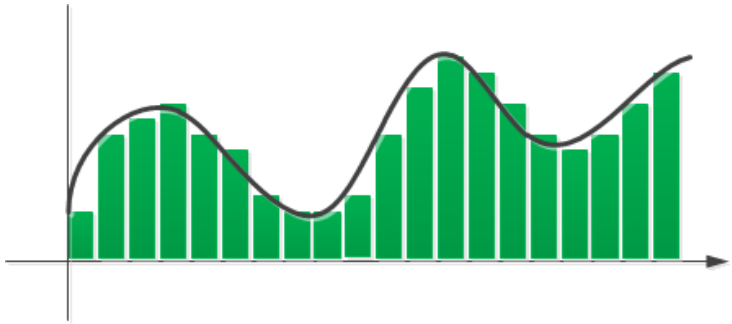
Durian Song
CD quality

Digital storage mode is convenient for long-term stable storage of data.

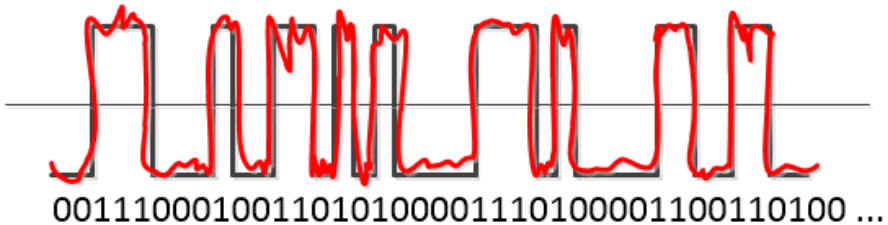
Main characteristics of digital signal

Easy transmission





Noise interference of analog signal transmission



Noise interference in digital signal transmission

In digital transmission, the signal amplitude is encoded by binary 0 and 1. Compared with analog signals, it has strong anti-interference ability.

Main characteristics of digital signal

Easy transmission

Strong anti-interference ability

For example: china national radio's opening recording

Analog signal transmission

Digital signal transmission

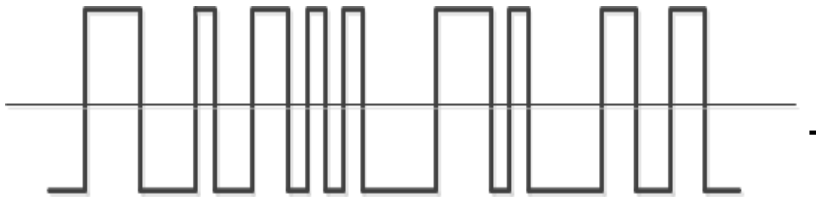
Digital transmission effectively resists channel interference and has good sound quality.

Main characteristics of digital signal

Easy to handle

Signal encryption

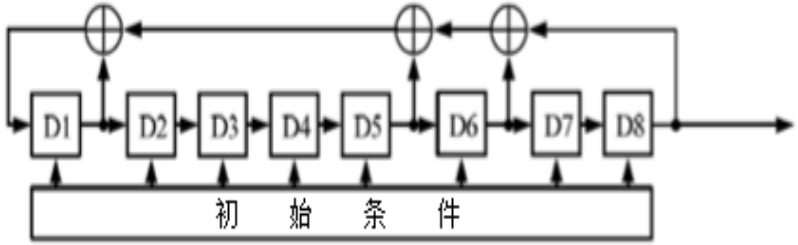
voice signal



0011100010011010100001110100001100110100 ...

Encrypted voice

pseudo-random code

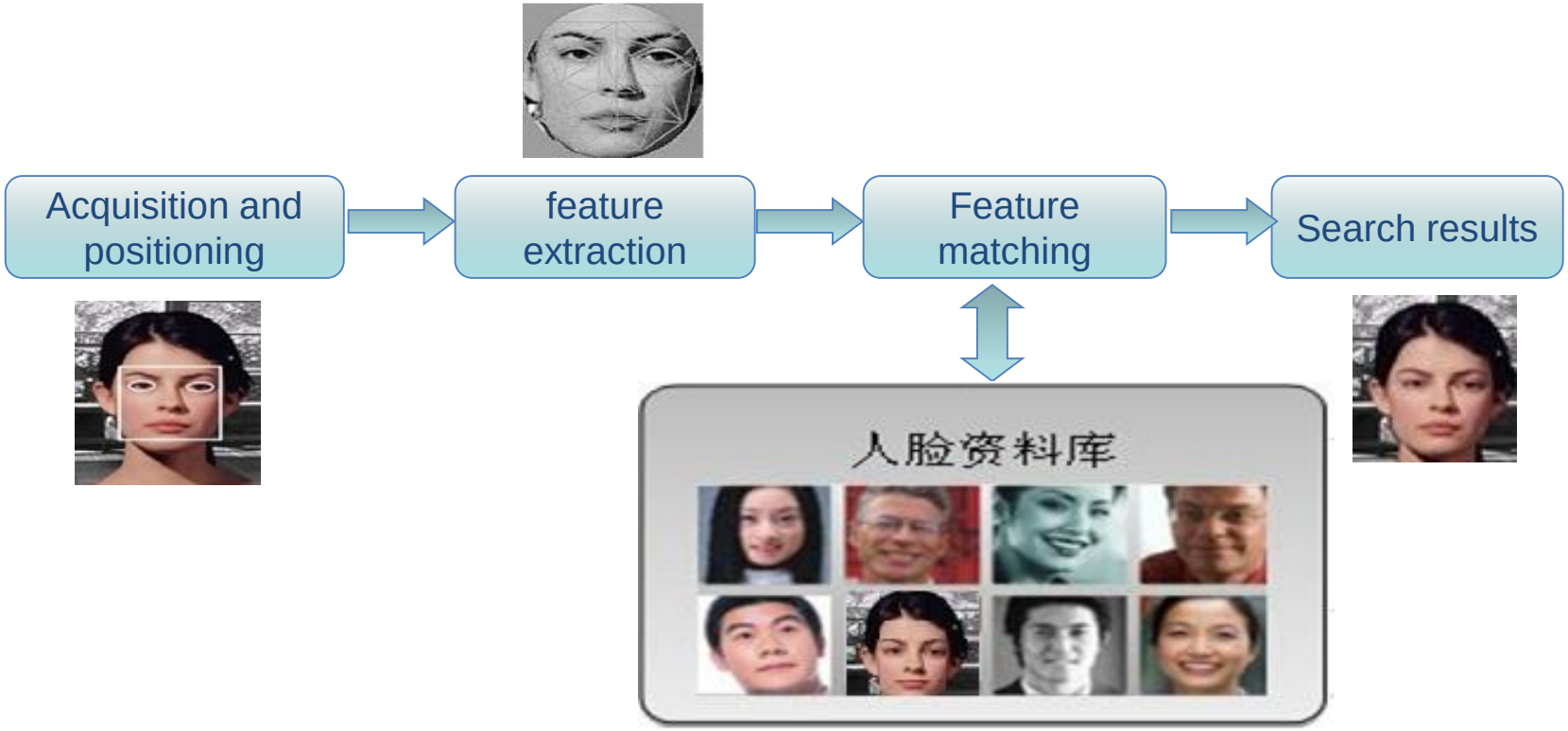


Good confidentiality

Main characteristics of digital signal

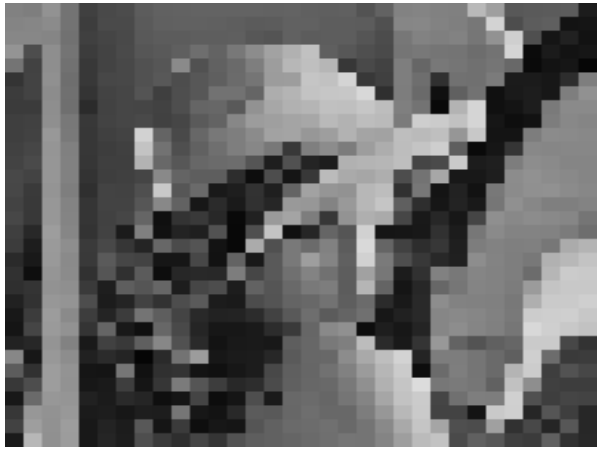
Easy to handle

Recognition retrieval



3. How to sample signals?

Influence of sampling interval on image



3. How to sample signals?

Influence of sampling frequency on speech signal

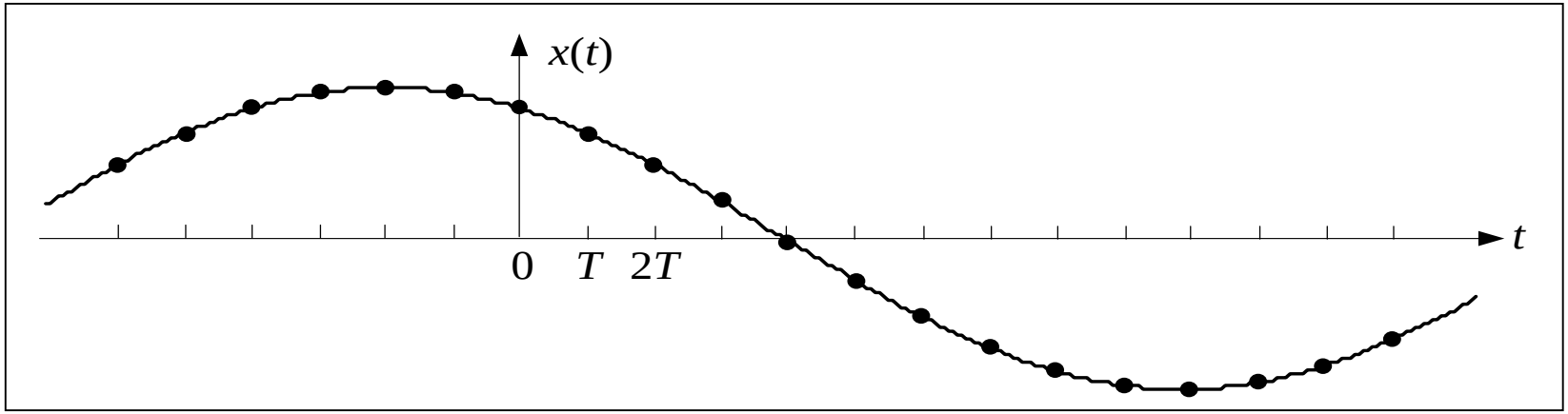
fsam=44,100 Hz

fsam=22,050 Hz

fsam=11,025 Hz

fsam=5,512 Hz

3. How to sample signals?

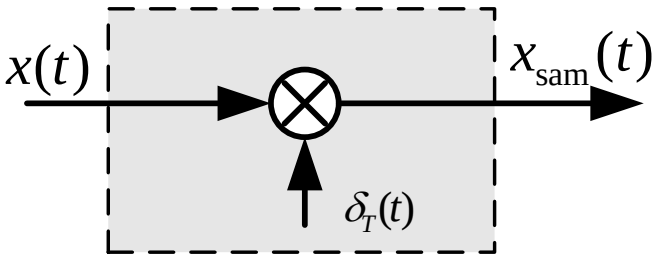


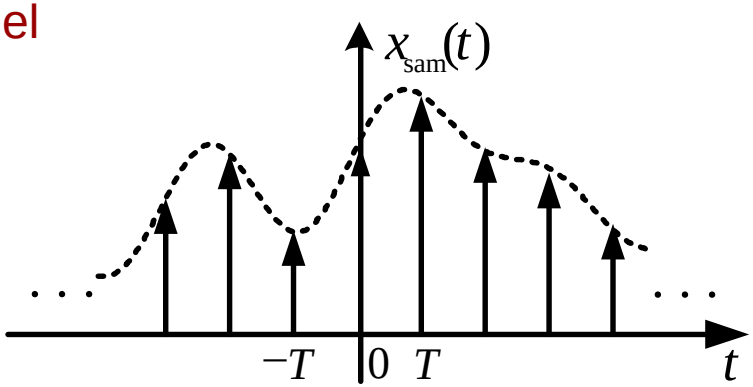
$$x[k] = x(t)\big|_{t=kT}$$

How to select the sampling interval t?

4. Theoretical analysis of signal sampling in time domain

Traditional model

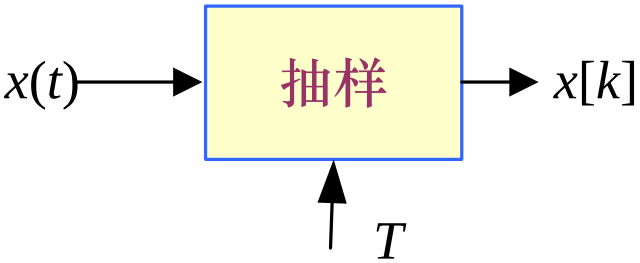


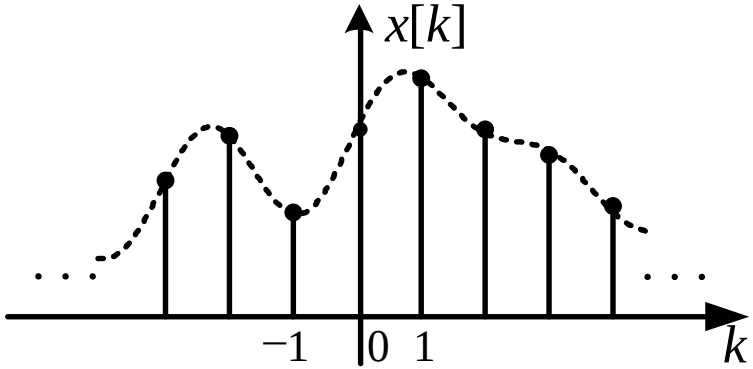


Sampling theorem proving model

Both input and output are continuous-time signals.

New model





The input is a continuous-time signal and the output is a discrete-time signal.

. Theoretical analysis of signal sampling in time domain

$$\begin{tikzcd}
x(t) \arrow[r, "t=kT"] \arrow[d, leftrightarrow] & x[k] \arrow[l, "?"] \arrow[d, leftrightarrow] \\
X(\mathrm{j}\omega) & X(\mathrm{e}^{\mathrm{j}\Omega}) \arrow[l]
\end{tikzcd}$$

$$x[k] = x(t)\big|_{t=kT}$$

$$(\Omega = \omega T)$$

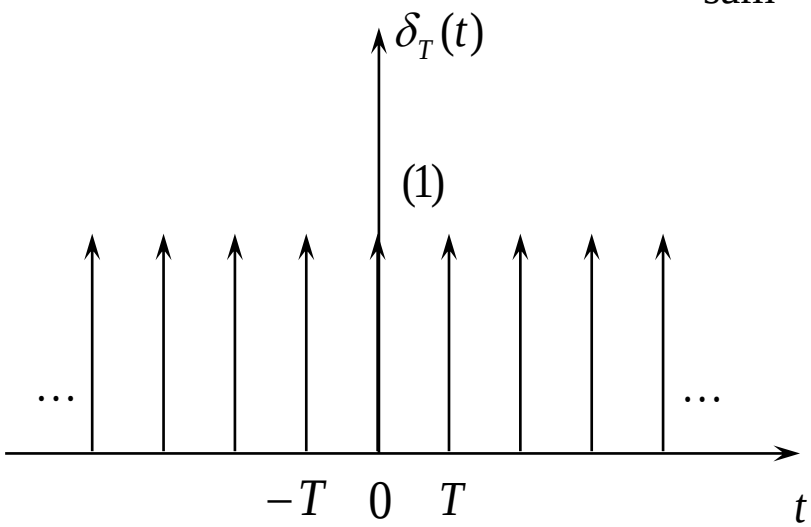
The frequency spectrum of continuous signal $x(t)$ is $X(\mathrm{j}\omega)$,

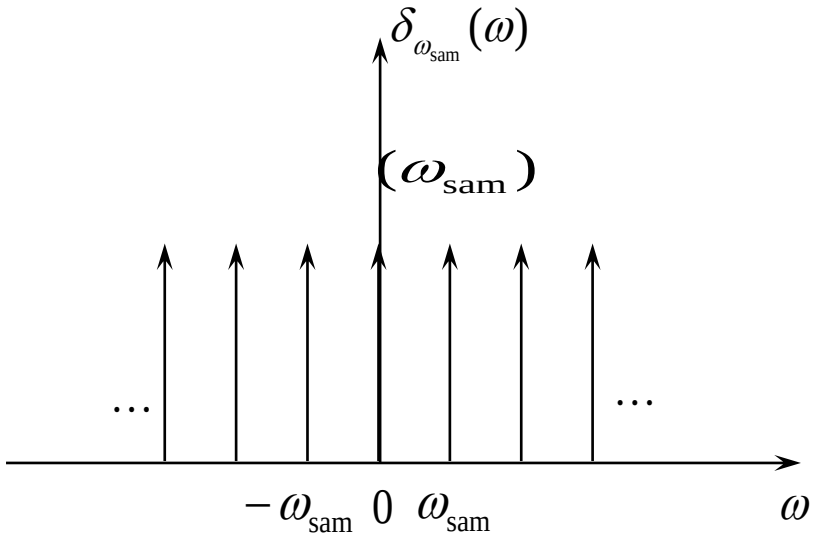
The spectrum of discrete sequence $x[k]$ is $X(\mathrm{e}^{\mathrm{j}\Omega})$.

4. Theoretical analysis of signal sampling in time domain

$$\delta_T(t) = \sum_{k=-\infty}^{+\infty} \delta(t - kT) \qquad \delta_{\omega_{\text{sam}}}(\omega) = \omega_{\text{sam}} \sum_{n=-\infty}^{+\infty} \delta(\omega - n\omega_{\text{sam}})$$

$$\omega_{\text{sam}} = 2\pi / T$$

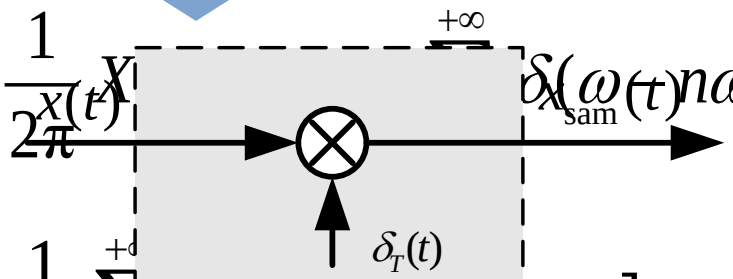




4. Theoretical analysis of signal sampling in time domain

$$x_{\text{sam}}(t) = \underline{x(t)\delta_T(t)} = \sum_{k=-\infty}^{+\infty} \underline{x(kT)\delta(t-kT)}$$

$$X_{\text{sam}}(\mathrm{j}\omega) = \frac{1}{2\pi} X_{\text{sam}}(t) * \delta(\omega - n\omega_{\text{sam}})$$



$$= \frac{1}{T} \sum_{n=-\infty}^{+\infty} X\left[\mathrm{j}(\omega - n\omega_{\text{sam}})\right]$$

Sampling theorem proving model

$$X_{\text{sam}}(\mathrm{j}\omega) = \sum_{k=-\infty}^{+\infty} x(kT)\mathrm{e}^{-\mathrm{j}k\omega T} = \sum_{k=-\infty}^{+\infty} x[k]\mathrm{e}^{-\mathrm{j}k\Omega} = X(\mathrm{e}^{\mathrm{j}\Omega})$$

$$X(\mathrm{e}^{\mathrm{j}\Omega}) = \frac{1}{T} \sum_{n=-\infty}^{+\infty} X\left[\mathrm{j}(\omega - n\omega_{\text{sam}})\right] \qquad (\Omega = \omega T)$$

4. Theoretical analysis of signal sampling in time domain

If the frequency spectrum of continuous signal $x(t)$ is $X(\mathrm{j}\omega)$ and the frequency spectrum of discrete sequence $x[k]$ is $X(\mathrm{e}^{\mathrm{j}\Omega})$ and exists.

$$x[k] = x(t)\big|_{t=kT}$$

Then there is

$$X(\mathrm{e}^{\mathrm{j}\Omega}) = \frac{1}{T}\sum_{n=-\infty}^{+\infty} X\left[\mathrm{j}(\omega - n\omega_{\mathrm{sam}})\right] \qquad (\Omega = \omega T)$$

The essence of sampling theorem: the discretization of signal in time domain leads to the periodicity of its frequency domain.

Where: T is the sampling interval and $\omega_{\mathrm{sam}}=2\pi/T$ is the sampling angular frequency.

4. Theoretical analysis of signal sampling in time domain

Band-limited signal

If the highest angular frequency of the band-limited signal $x(t)$ is ω_m, the signal $x(t)$ can be uniquely represented by the sampling values with equal intervals T under certain conditions.

The sampling interval t shall meet the following requirements:

$$T \le \pi/\omega_m = 1/(2f_m)$$

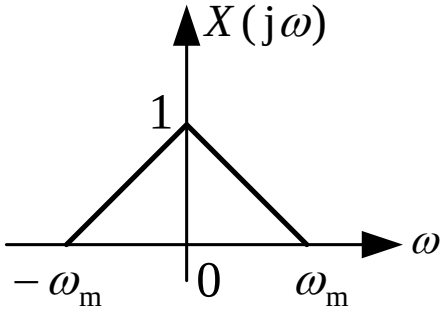
$$f_{sam} \ge 2f_m \quad (\text{or } \omega_{sam} \ge 2\omega_m)$$

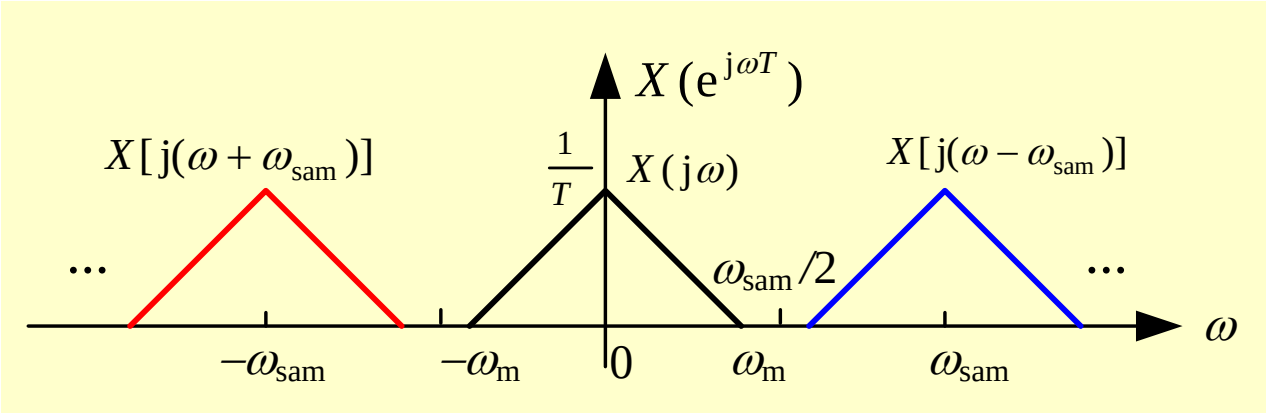
$f_{sam} = 2f_m$ is the minimum sampling frequency, which is called Nyquist Rate.

4. Theoretical analysis of signal sampling in time domain

The relationship between the spectrum of discrete sequence $x[k]$ and sampling interval T.

$$\omega_{\text{sam}} > 2\omega_{\text{m}}$$

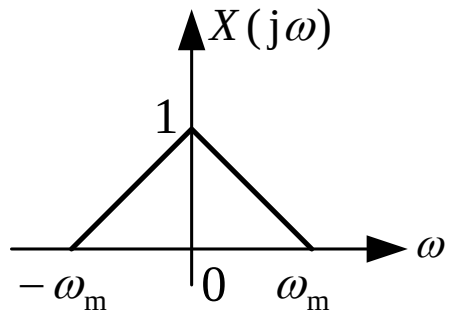


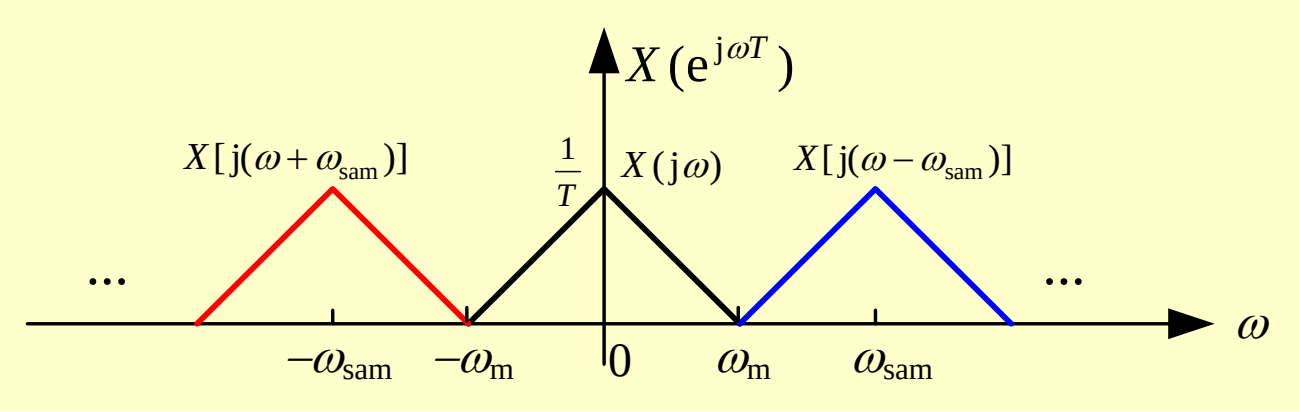


4. Theoretical analysis of signal sampling in time domain

The relationship between the spectrum of discrete sequence $x[k]$ and sampling interval T.

$$\omega_{\text{sam}} = 2\omega_{\text{m}}$$

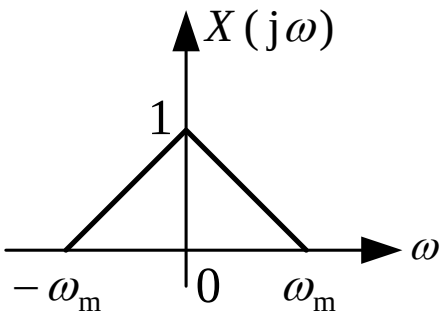


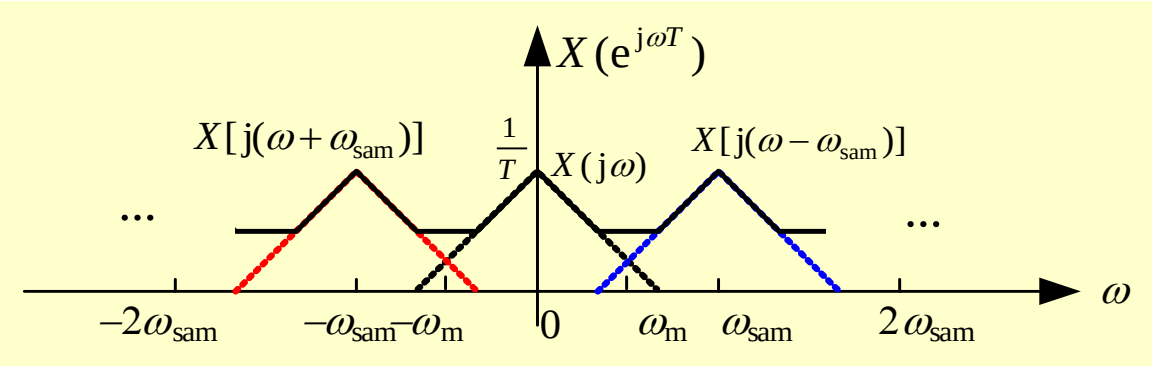


4. Theoretical analysis of signal sampling in time domain

The relationship between the spectrum of discrete sequence $x[k]$ and sampling interval T.

$$\omega_{\text{sam}} < 2\omega_{\text{m}}$$





Aliasing

4. Theoretical analysis of signal sampling in time domain

Ex.: It is known that the highest frequency of real signal $x(t)$ is f_m (Hz), and the minimum sampling frequency for sampling each signal $x(2t)$, $x(t)*x(2t)$ and $x(t)\cdot x(2t)$ is calculated.

Solution:
According to the correspondence between signal time domain and frequency domain and the sampling theorem, it is obtained that:

When sampling signal $x(2t)$, the minimum sampling frequency is $4f_m$ (Hz);

When sampling $x(t)*x(2t)$, the minimum sampling frequency is $2f_m$(Hz);

When sampling $x(t)\cdot x(2t)$, the minimum sampling frequency is $6f_m$(Hz).

4. Theoretical analysis of signal sampling in time domain

On time domain sampling of signals, ...

(1) Does the non-band-limited signal sampling undistorted condition also have to satisfy $f_{\mathrm{sam}} \geq 2f_{\mathrm{m}}$?

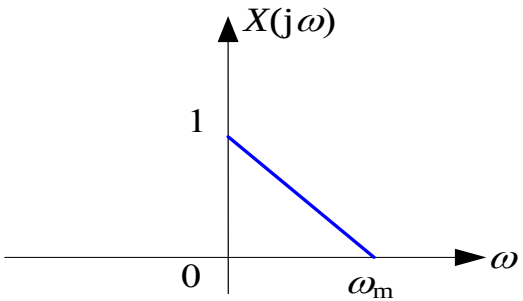
(2) When sampling continuous band-limited signals, only the sampling rate $f_{\mathrm{sam}} \geq 2f_{\mathrm{m}}$ is needed. In engineering application, why is the sampling rate always $f_{\mathrm{sam}} \geq (3\sim5)f_{\mathrm{m}}$?

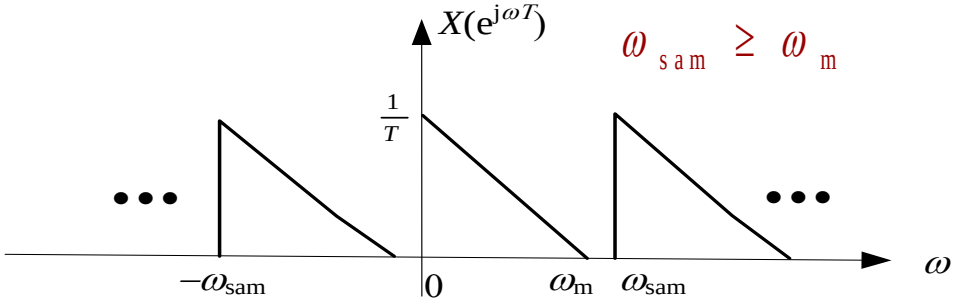
(3) If the highest frequency of continuous-time signal $x(t)$ is unknown, how to determine the sampling interval T of signal?

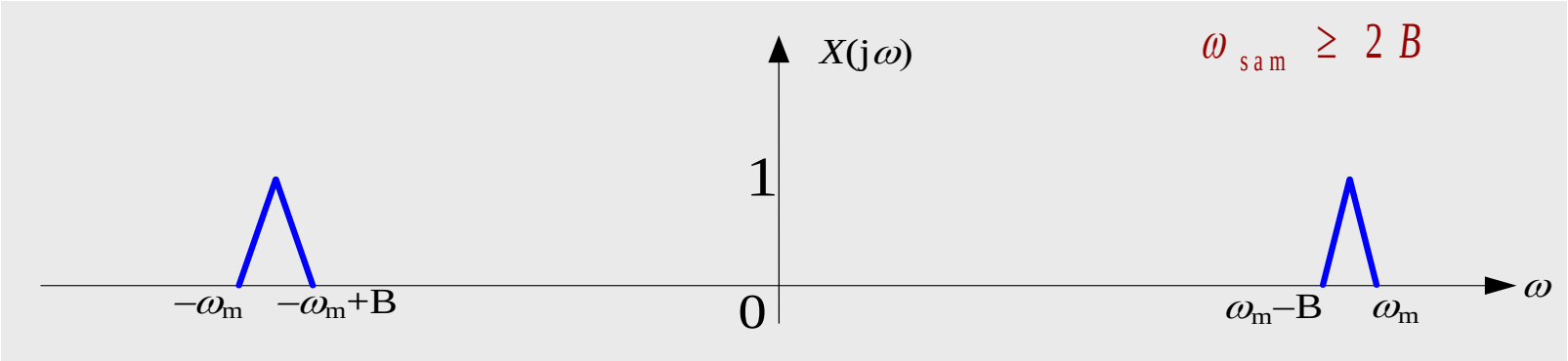
(4) If the sampling rate is too high, how to reduce the sampling rate of the sampled signal?

. Theoretical analysis of signal sampling in time domain

Sampling problem of single sideband signal and narrowband high frequency signal



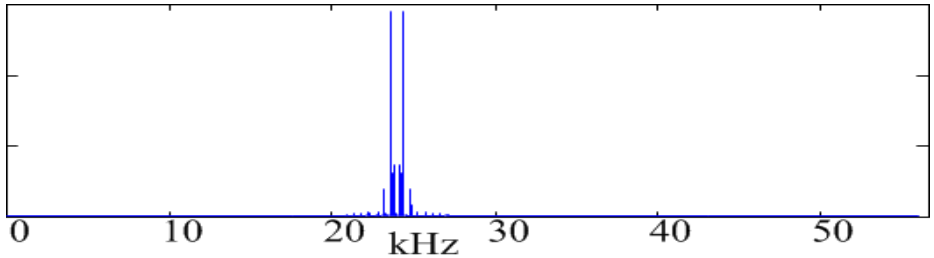




$$\omega_{\mathrm{m}} = 1000\mathrm{k\ rad/s}, \quad B = 8\mathrm{k\ rad/s}$$

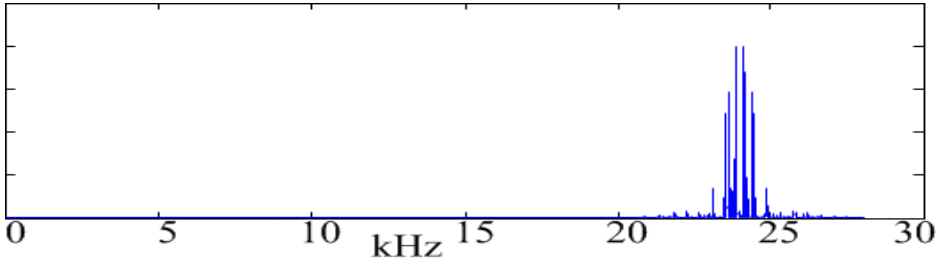
4. Theoretical analysis of signal sampling in time domain

Sampling of narrowband high frequency signals



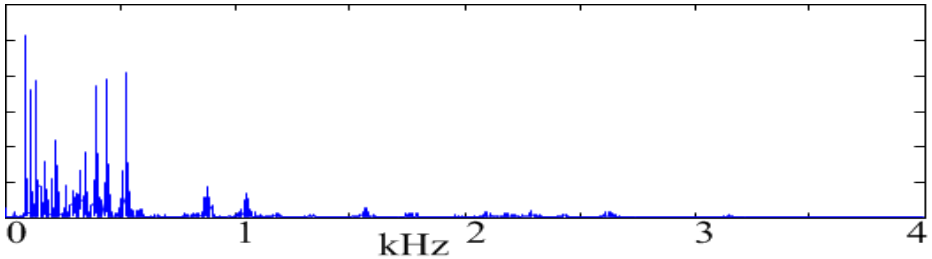
The center frequency is 24kHz and the bandwidth is 8kHz.

Demodulated speech signal



f_{sam}=56kHz sampled spectrum.

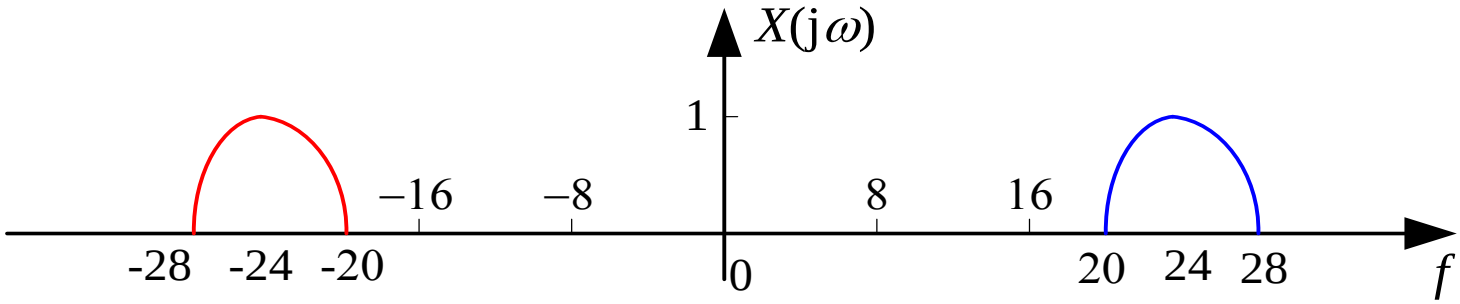
Demodulated speech signal



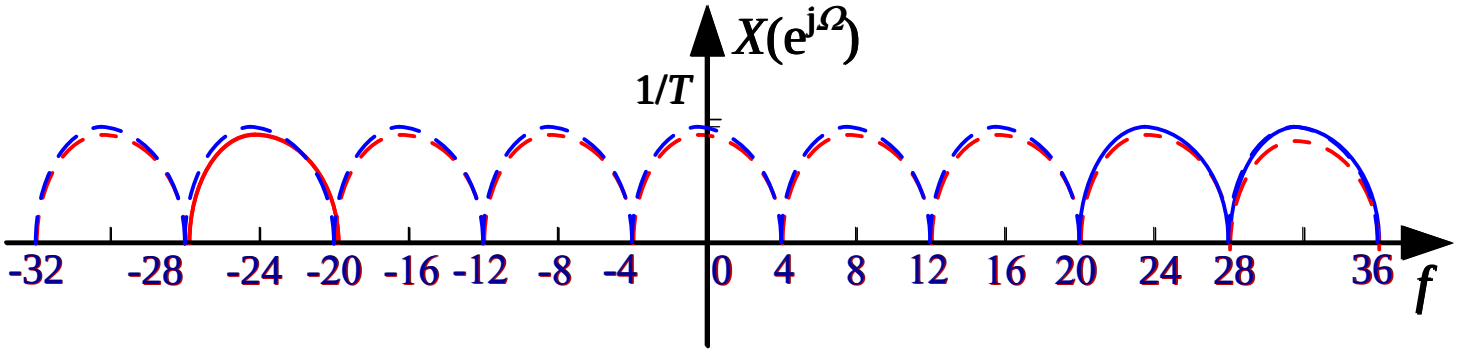
f_{sam}=8kHz sampled spectrum.

Sampled speech signal (not demodulated)

4. Theoretical analysis of signal sampling in time domain

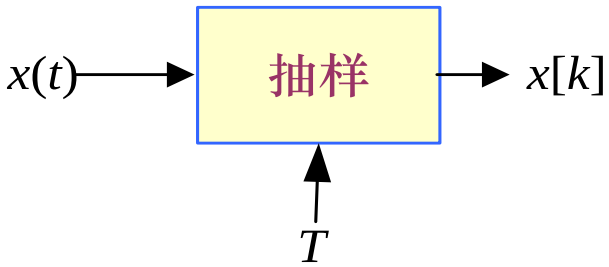


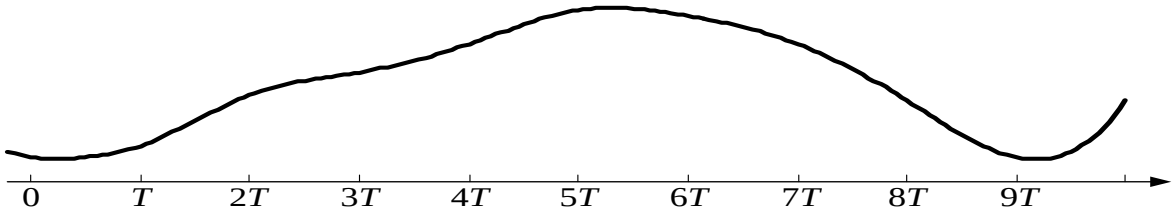
$f_m = 28\ kHz$



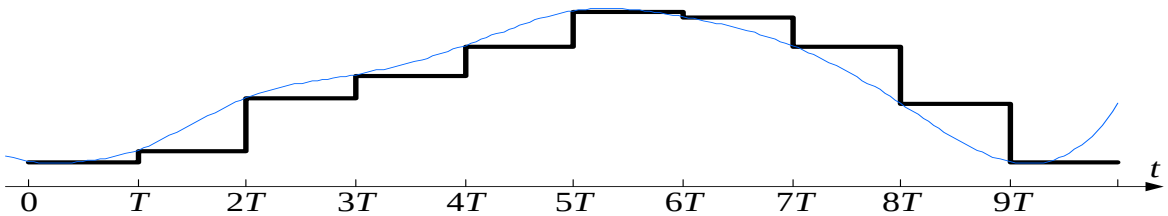
$f_{sam} = 8\ kHz$

5. Engineering application of sampling theorem



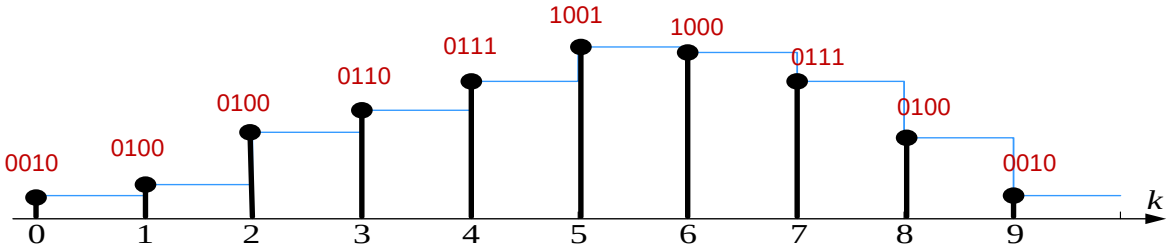


Continuous time signal



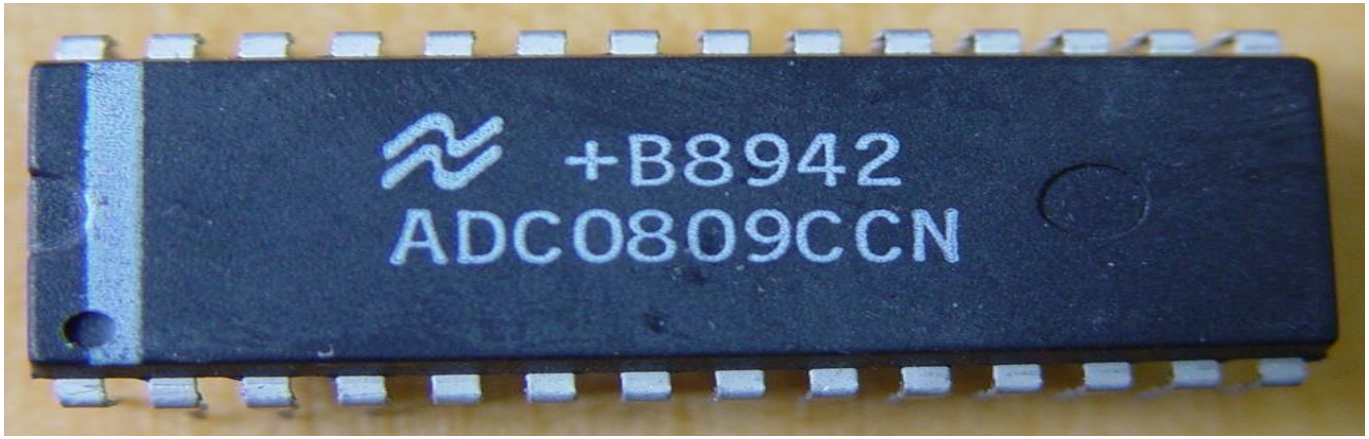
Sample and hold

Quantization coding



discrete time signals Digital signal

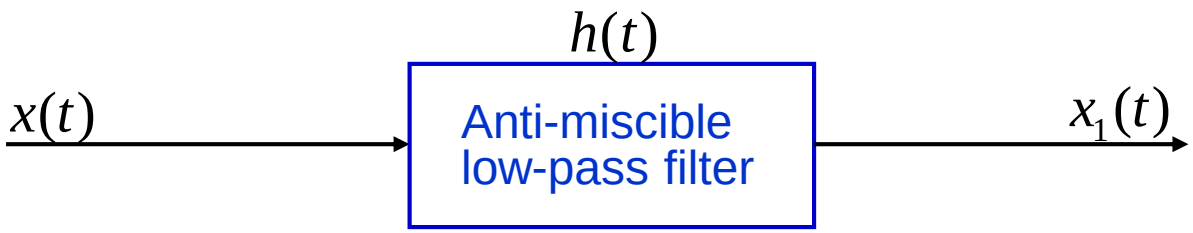
5. Engineering application of sampling theorem

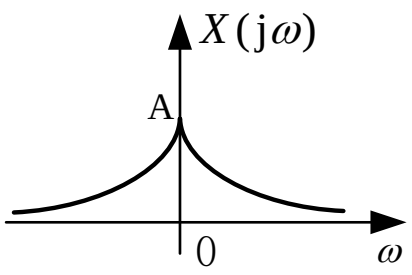


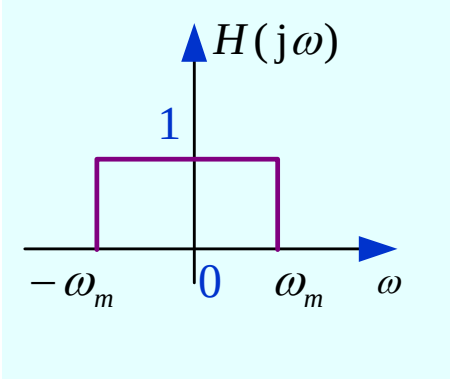
Sampling interval (period)	T (s)
Sampling angular frequency	$\omega_{sam}=2\pi/T$ (rad/s)
Sampling frequency	$f_{sam}=1/T$(Hz)

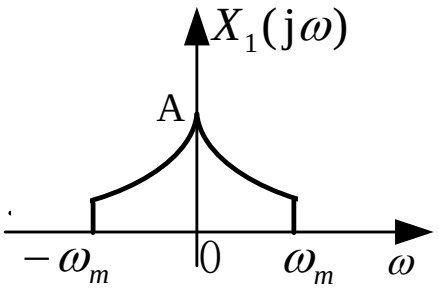
5. Engineering application of sampling theorem

Many practical engineering signals do not meet the band-limited conditions.



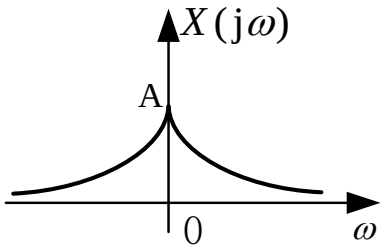


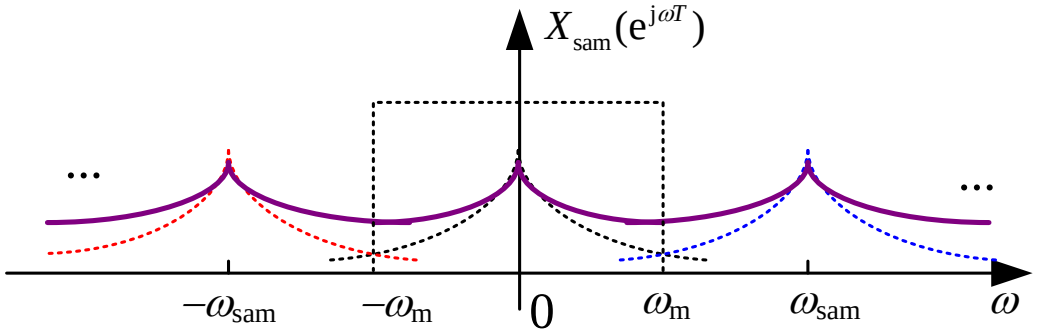


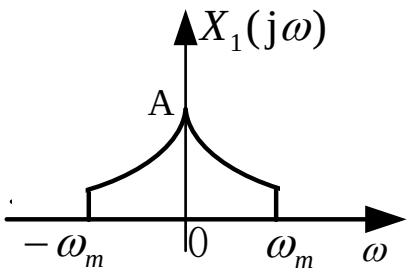


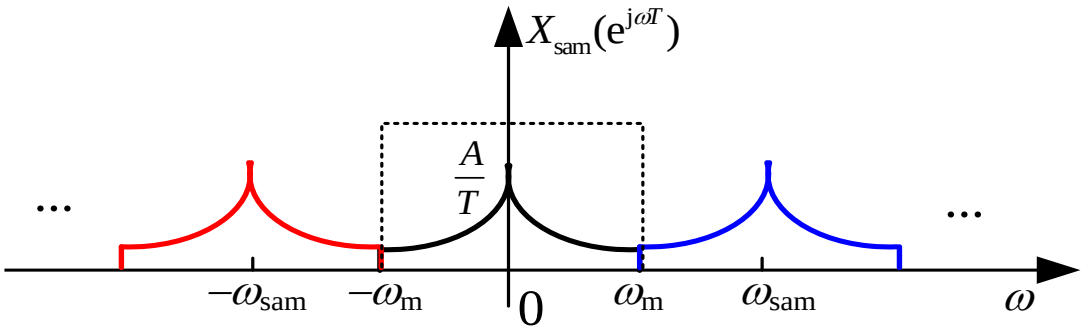
5. Engineering application of sampling theorem

Comparison between aliasing error and truncation error









5. Engineering application of sampling theorem

Anti-aliasing filtering of speech signal

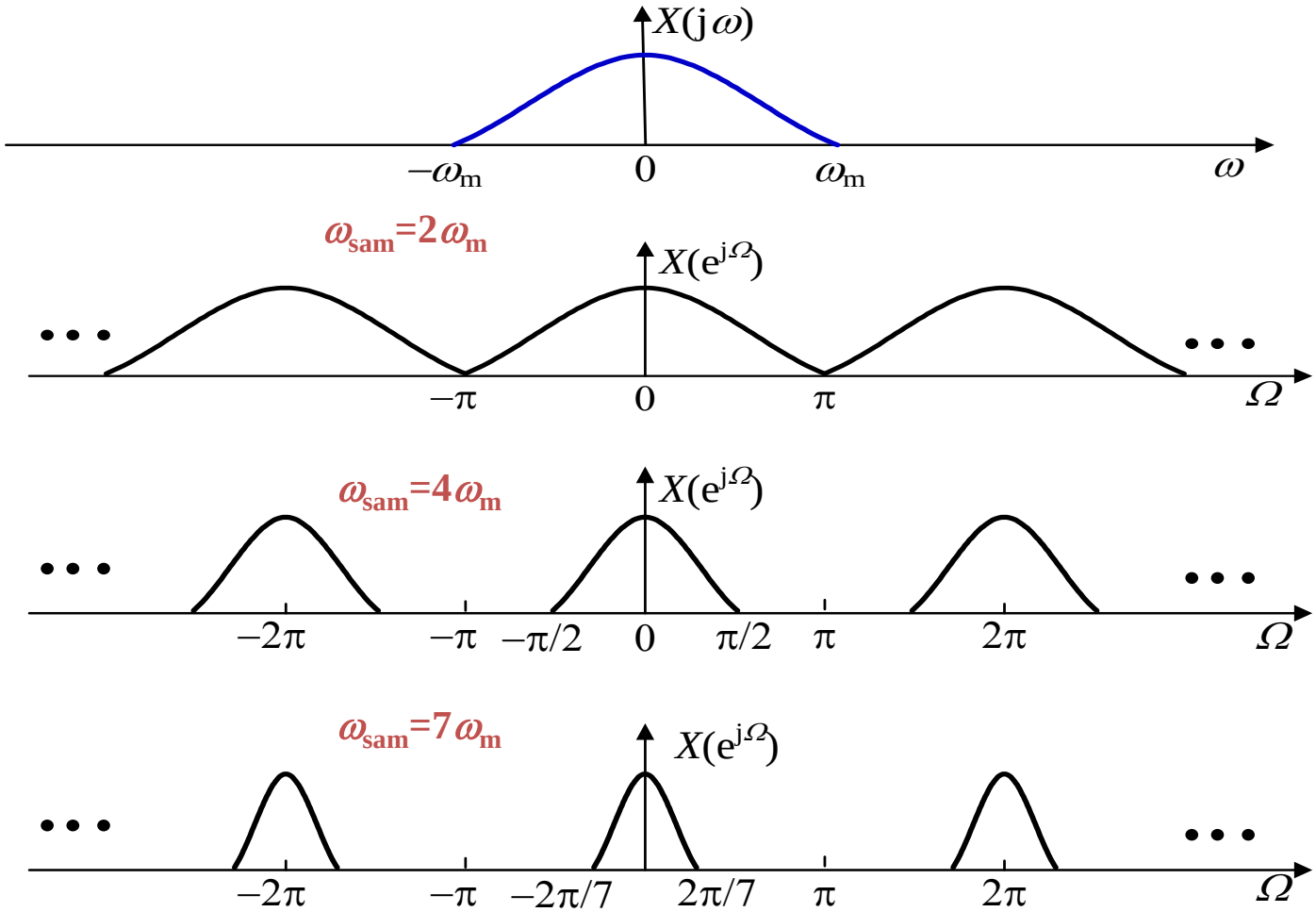
Sampling frequency f_{sam} =44,100 Hz.

Sampling frequency f_{sam} =5,512 Hz.

Sampling frequency f_{sam} =5,512 Hz.
Anti-aliasing filtering is performed on the signal before sampling.

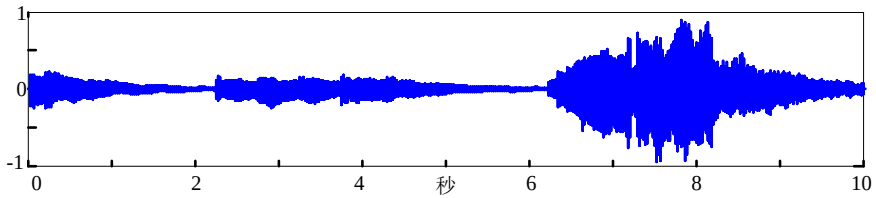
5. Engineering application of sampling theorem

Signal sampling at the highest frequency of unknown signal

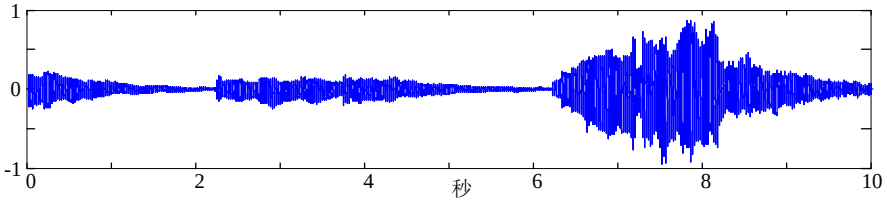


5. Engineering application of sampling theorem

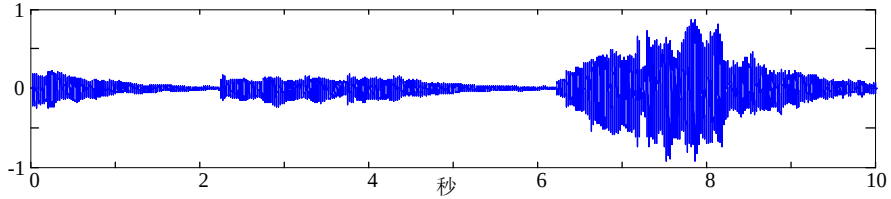
Signal sampling at the highest frequency of unknown signal



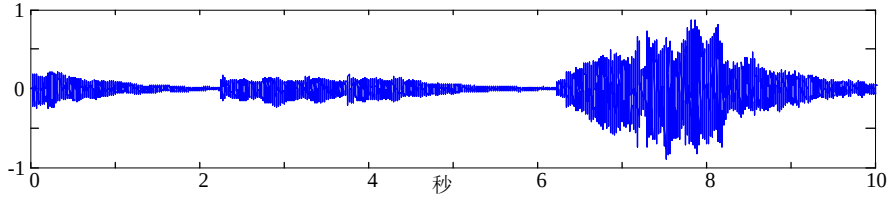
$f_{sam} = 44{,}100\ Hz$



$f_{sam} = 22{,}050\ Hz$



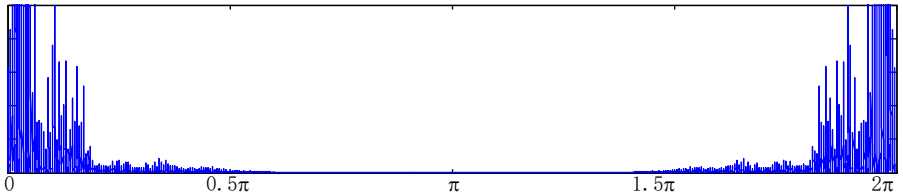
$f_{sam} = 11{,}025\ Hz$



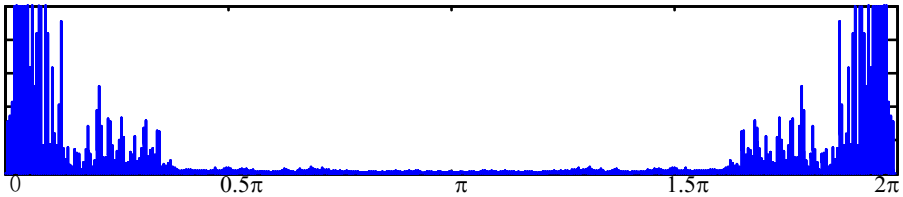
$f_{sam} = 5{,}512\ Hz$

5. Engineering application of sampling theorem

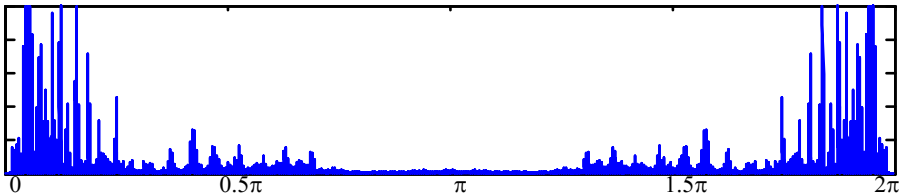
Signal sampling at the highest frequency of unknown signal



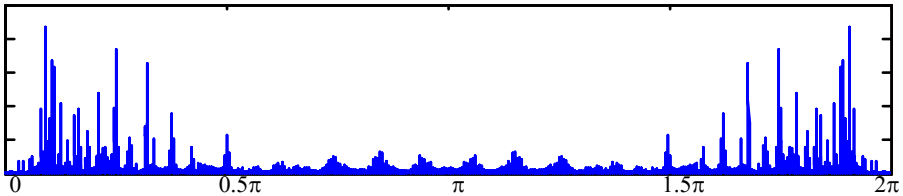
f_{sam} =44100 Hz



f_{sam} =22050 Hz



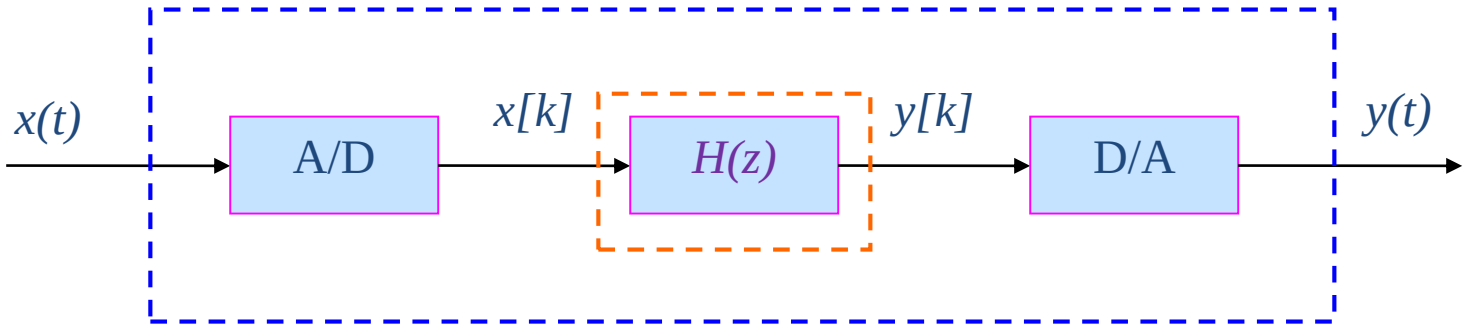
f_{sam} =11025 Hz



f_{sam} =5512 Hz

5. Engineering application of sampling theorem

Processing continuous-time signals with discrete systems

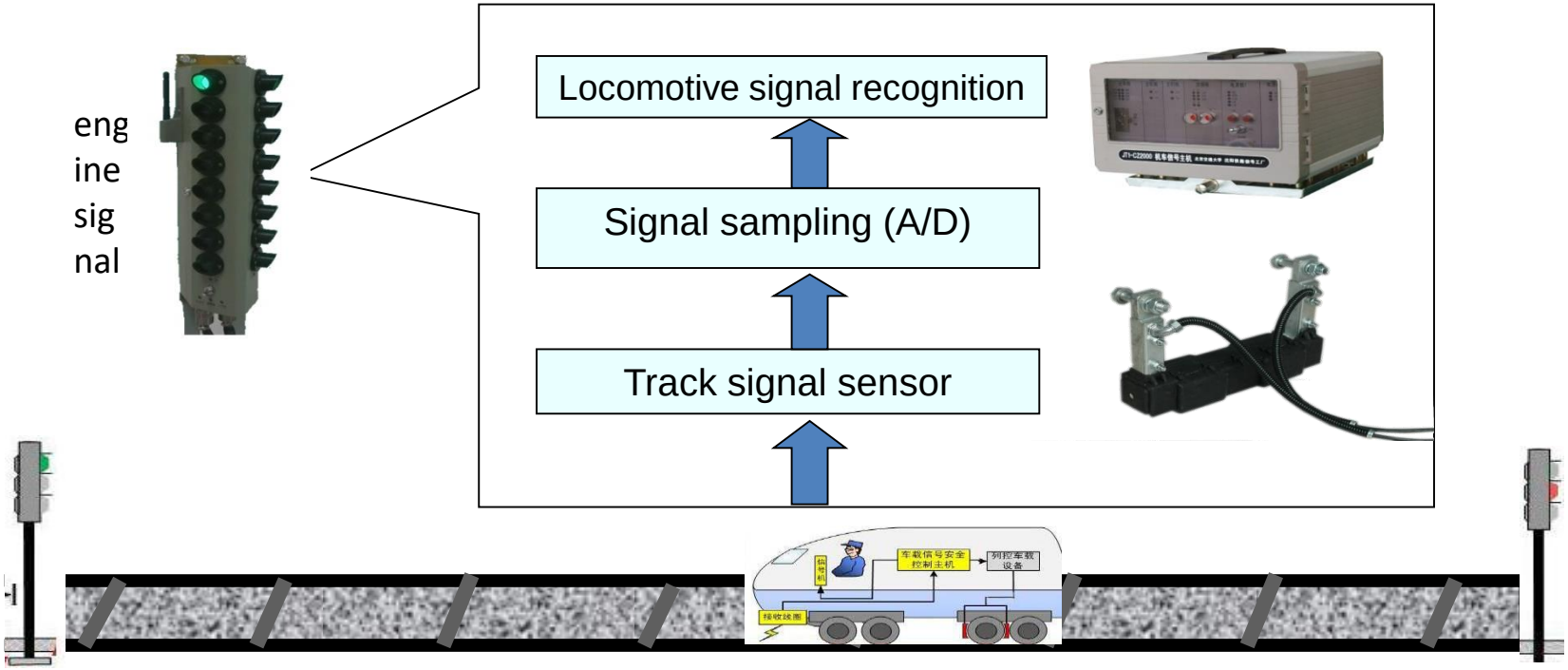


Railway control signal recognition

Biomedical signal processing

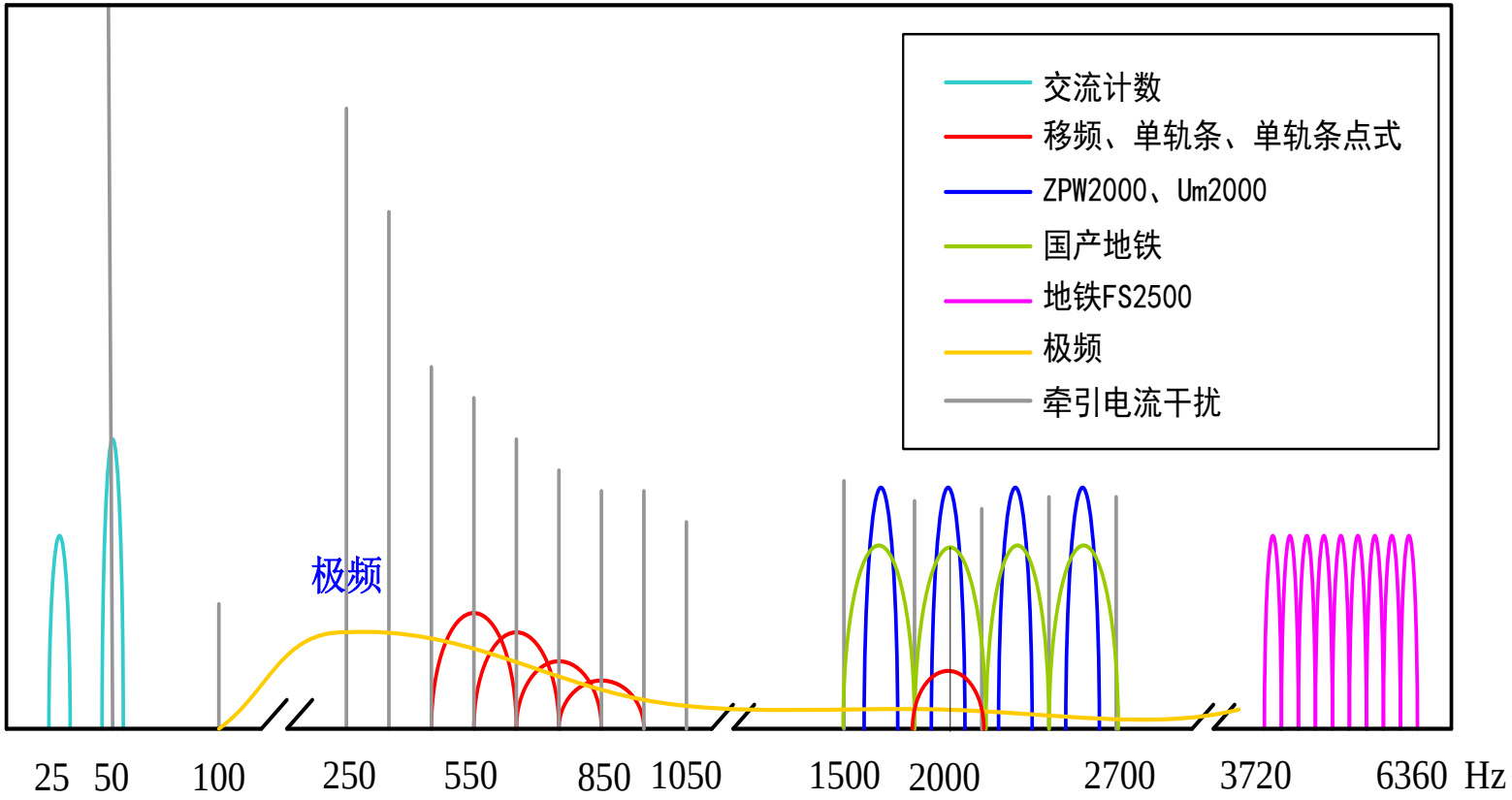
5. Engineering application of sampling theorem

Railway control signal recognition



5. Engineering application of sampling theorem

Railway control signal recognition

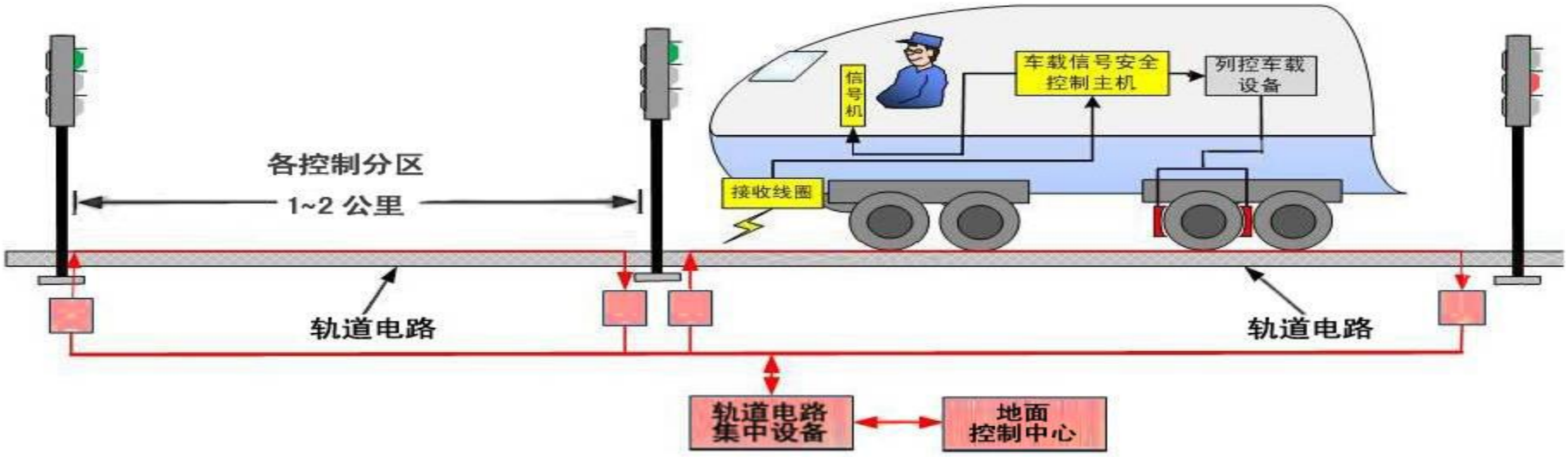


Spectrum of multi-system train control signal

5. Engineering application of sampling theorem

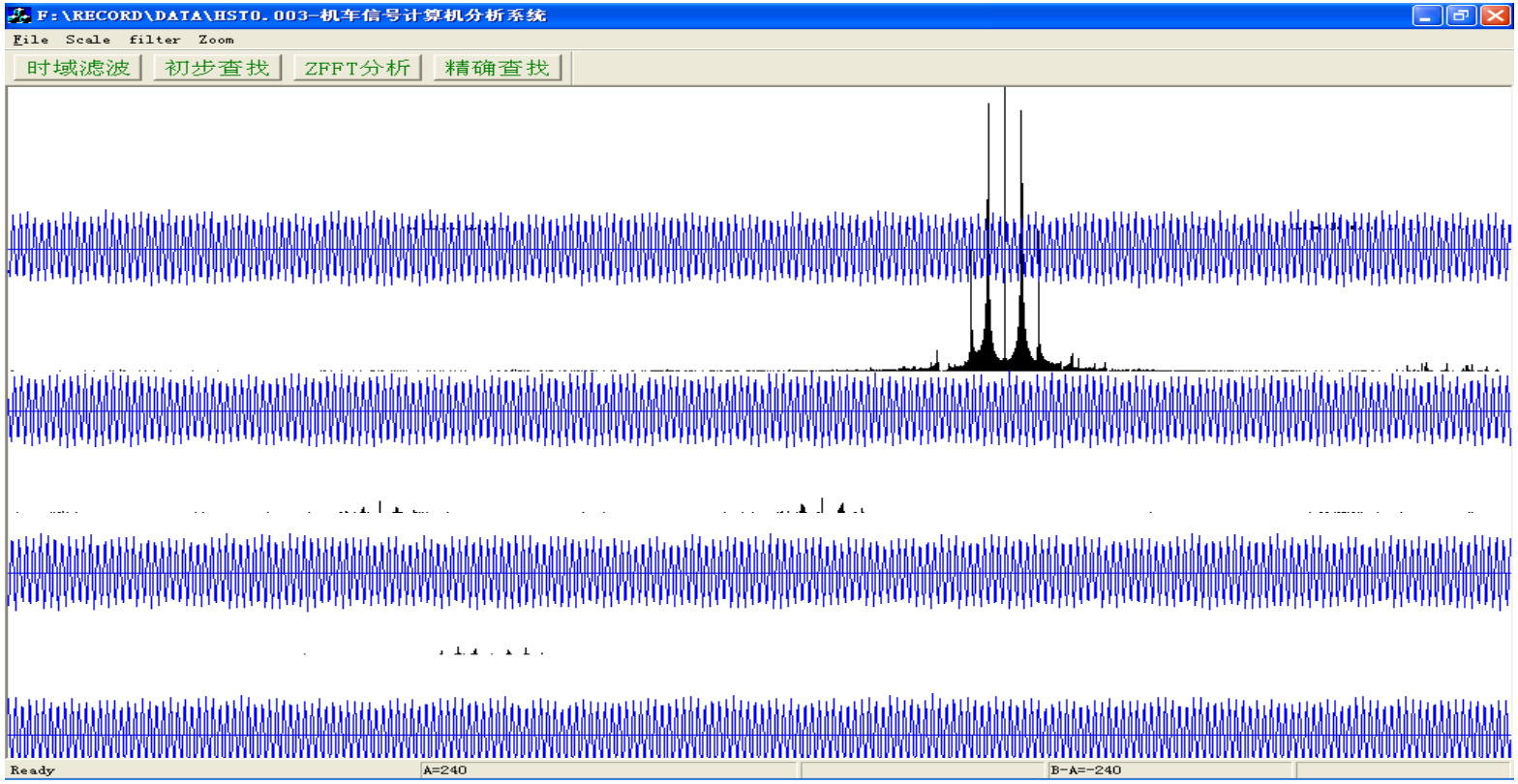
Due to the technical limitations of safety and reliability, the traditional on-board signal system can only be used as an auxiliary signal, and drivers must watch the ground signal to drive the train.

It is internationally recognized that the safety of trains with a speed of more than 160km/h or a high density can no longer be guaranteed by the driver's observation of the ground signal, but the on-board signal must be used as the main signal to control the train.



5. Engineering application of sampling theorem

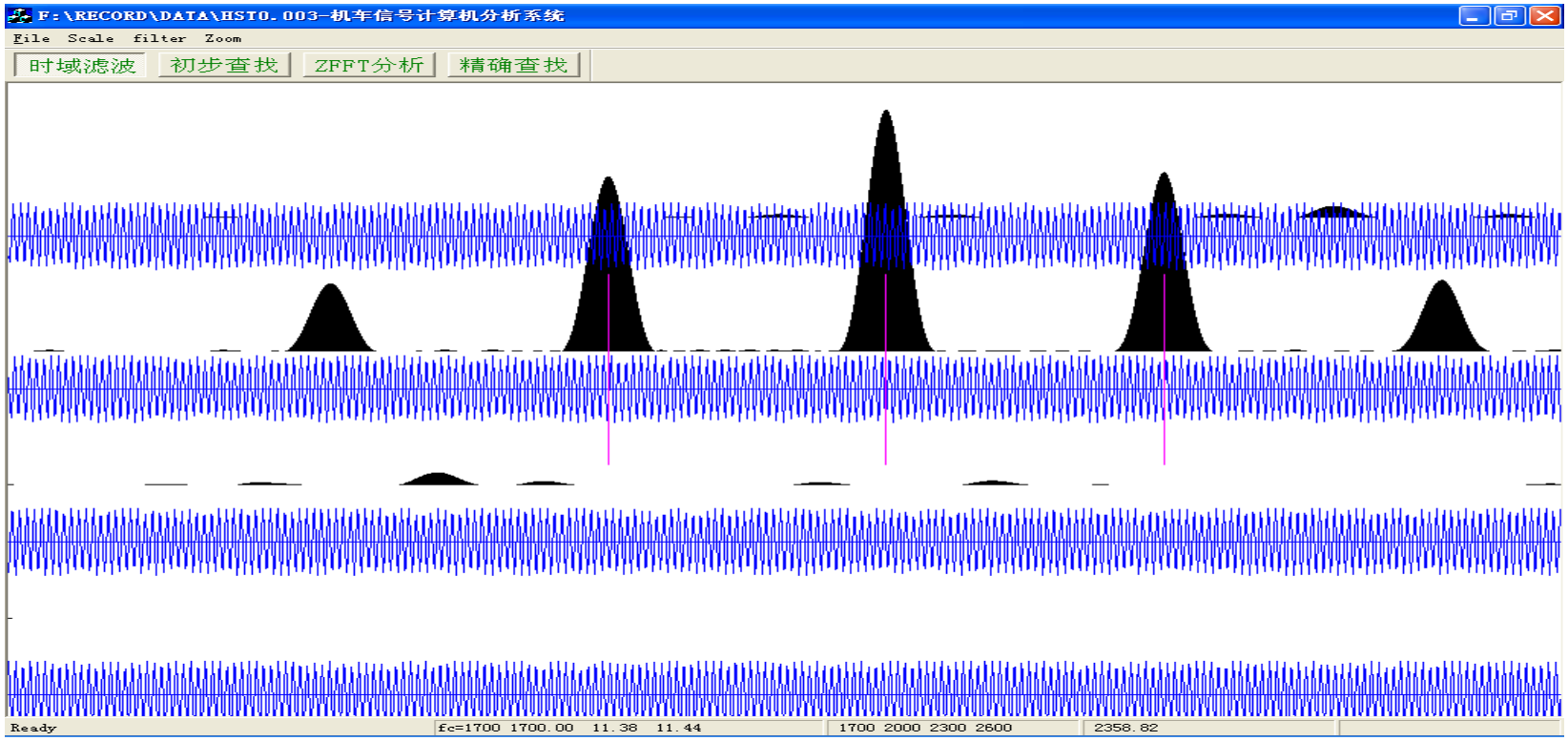
Railway control signal recognition



Time domain waveform and frequency spectrum of railway control signal

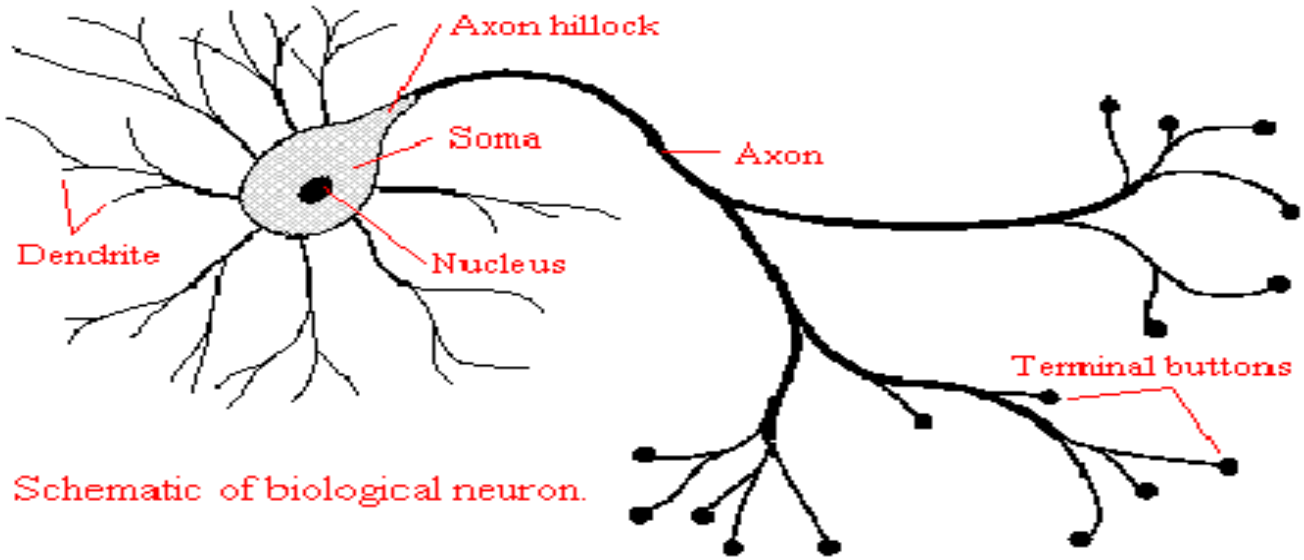
5. Engineering application of sampling theorem

Railway control signal recognition



Spectrum analysis of railway control signal

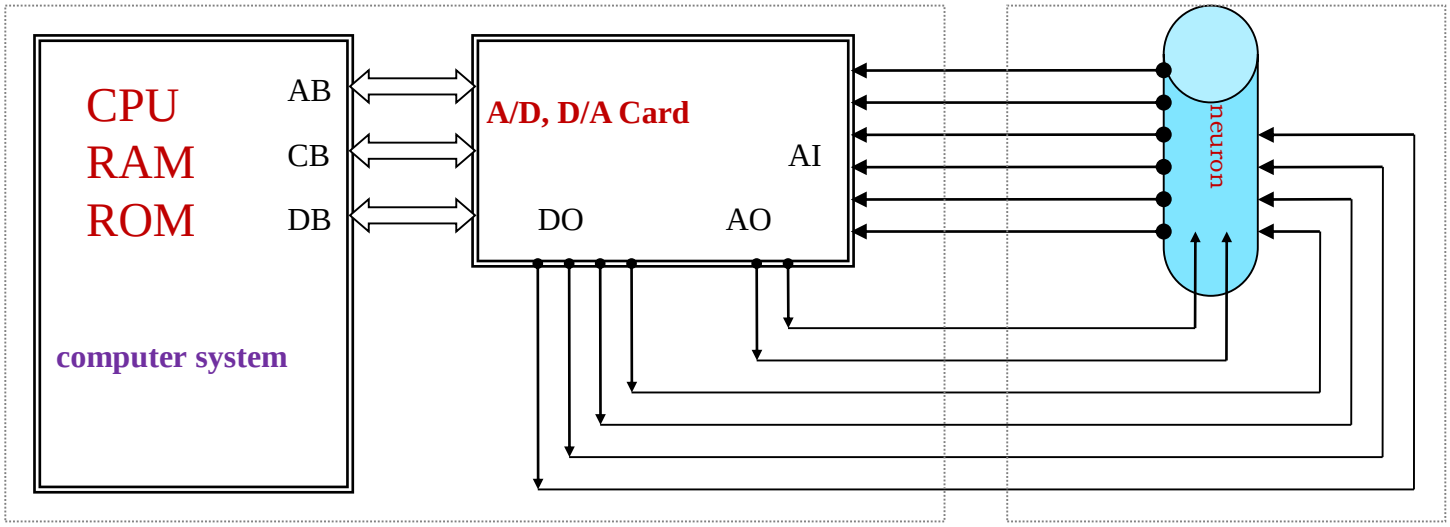
Biomedical signal processing



Structural diagram of biological nerve cells (elements)

5. Engineering application of sampling theorem

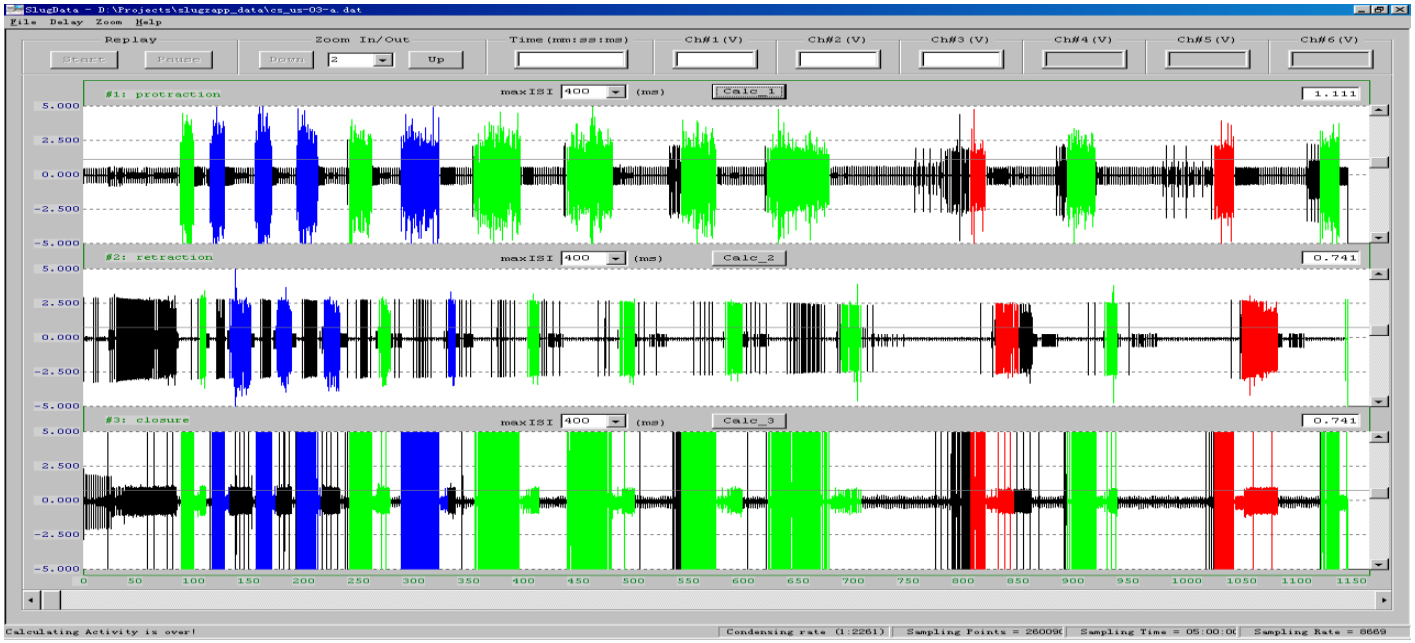
Biomedical signal processing



Composition block diagram of biological signal acquisition system

5. Engineering application of sampling theorem

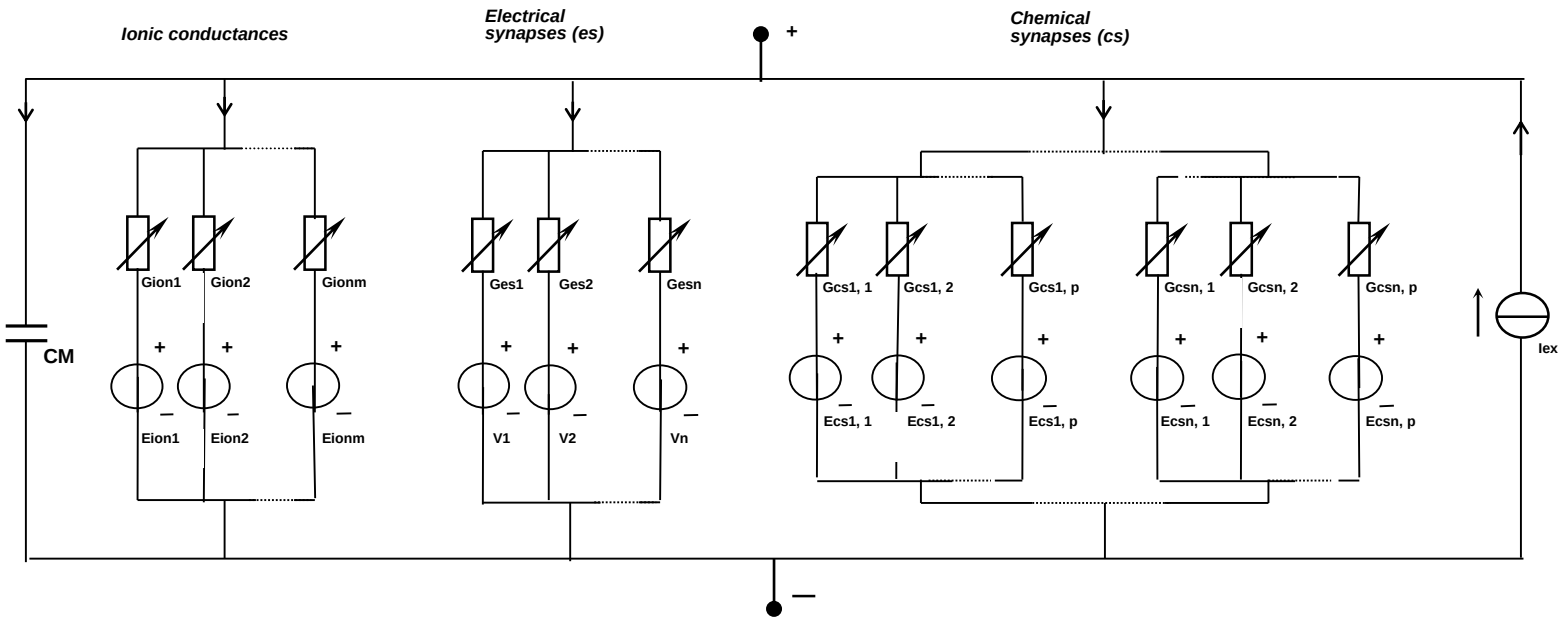
Biomedical signal processing



Pattern recognition of collected biological signals

5. Engineering application of sampling theorem

Biomedical signal processing



Neuron equivalent circuit

Thanks

Some of the materials cited in this course are the teaching accumulation of the main lecturer for many years, which come from a variety of media and the exchanges of colleagues, peers and friends, and it is difficult to indicate the source one by one, so I hereby explain and thank you!